

Rick (Frederick) M. Segrest

Senior Software Engineer & Subject Matter Expert

Contact information available on request

PROFESSIONAL SUMMARY

Senior Software Engineer with 20+ years of experience specializing in UI/UX design, full-stack web applications, and data visualization for complex scientific, aerospace, and defense systems. Currently sole developer on a Microsoft Power Apps coordination and estimating tool for NASA's Materials & Processes division, projected to save six figures annually. Proven expertise in React, TypeScript, Python, and modern DevOps practices, with practical fluency in agentic coding harnesses and multimodal AI tooling. Multiple innovation awards and group achievement honors from NASA and prime contractors.

CORE KNOWLEDGE & EXPERTISE

Extensive experience engineering user-friendly Graphical User Interfaces and front-ends, building Web Application GUIs in React, TypeScript/JavaScript, using Semantic-UI, IBM Carbon, and MaterialUI frameworks for web; also experienced with C++ with Qt, Java Swing & FX for legacy applications; and Swift + UIKit and React Native for Mobile development.

Visualization using multiple libraries, including Plotly, D3, NASA WorldWind, and custom functions. User Experience and GUI design and documentation using UI/UX, Figma, Adobe XD.

Built Backend APIs and systems using Python, Flask, Node.js, Socket.io & Web Sockets; Linux CLI and open source applications; and Database Management Systems including PostgreSQL, MSSQL, MongoDB and other NoSQL DBs. Microsoft Power Platform including Power Apps, Power Automate, and Dataverse.

Familiar with Software Development Processes such as Agile Scrum, Kanban, and Test-driven Development; DevOps using Docker, Git, Atlassian API, GitHub, and GitLab.

Practical experience selecting and integrating AI tooling across the modern stack — agentic coding harnesses (Hermes Agent, OpenClaw, Claude Code, and others) used as productivity multipliers alongside two decades of producing production software independently, with the judgment to pick the right harness, model, and deployment for a given task. Broad familiarity with commercial, cloud-hosted, and self-deployed local models. Integrating AI service APIs (proprietary and open-source LLMs, multimodal, and vision models) into applications to deploy cutting-edge features. Hands-on experience producing generative imagery, video, and audio with both local open-source and commercial models.

Statistical Analysis & Data Modeling: Applied advanced mathematical concepts and scientific tools (MATLAB, MiniTab, R, SciPy) to improve proprietary modeling and simulation systems for NASA and DoD.

Extensive writing experience, including technical writing, tutorials, presentations, and documentation for complex software and technical systems.

PROFESSIONAL EXPERIENCE

Amentum Space Exploration Group (formerly Jacobs) • Huntsville, AL

2019 – Present

Senior Software Engineer & Subject Matter Expert — GUI, UX, and Web Application Software Engineering

EM Materials & Processes Division • ED 04 Advanced Concepts Office • ES 50 ECLSS 4BCOS Digital Twin IRAD • ES 52 Flight Software Tools Team

EM Materials & Processes Division — Sole developer designing and building a Microsoft Power Apps coordination and estimating application for NASA's Materials & Processes (EM) division. The tool replaces fragmented, branch-level Excel estimating with a centralized intake-and-coordination system on Dataverse, projected to save the division six figures annually by consolidating multi-branch responses into a single coordinated EM estimate.

Designed a hybrid model-driven and canvas Power Apps architecture with a Work Request → Participating Org → Org Submission data model, a Business Process Flow driving the ticket lifecycle from intake through customer delivery, and canvas custom pages handling broadcast review and duplicate/merge workflows. Owning all architecture, data modeling, and integration decisions.

Led 5-person team for NASA Advanced Concepts Office (ACO) that delivered alpha version of **Dyreqt SaaS** in 8 weeks under FY surge, enabling non-Python users to adopt the tool and eliminating the need for week-long user training sessions led by busy simulation developers. Awarded *NASA MSFC Team Innovation Award* in Fall 2025 for “innovation and advancement of system-of-systems analysis methods applied to multi-element human exploration architectures.”

Created and taught two courses for the Jacobs MPP (Mentoring Peers Program): *Introduction to React Web Application Development* and *Streamlining DevOps*. Awarded 2023 TIPI Grant to author a 100-page internal tutorial book, facilitating knowledge transfer across NASA teams; completed in early 2024.

Extended NASA’s Sherlock Web Application front end for a big-data tool allowing users to consume, search, filter, and visualize SLS (Space Launch System) simulation data and real-world telemetry data.

Supported NASA IRAD to produce a Digital Twin Simulation of the ISS’s 4-bed CO₂ Scrubber physical system. Presented GUI innovations live and virtually at Jacobs’ MSFC Center-wide Program Review in September 2023. Team was awarded a *NASA Group Achievement Award* in 2024.

Built 3 Atlassian Jira/Confluence DevOps automation tools using Python Flask REST APIs with TS/JS React front-ends that eliminated manual reporting and automated repetitive tasks for 50+ users.

Primary developer in NASA IRAD effort to design, prototype, and now implement the NASA Software Engineering Requirements Tailoring Tool (SERTT).

Led an effort to recreate and modernize THOR, a legacy NASA application that queries a REST API to search and generate reports from the HOSC’s (Huntsville Operations Support Center) centralized library of raw ISS payload data and other artifacts.

Recognition

- Spot Bonus for support of multiple NASA teams, April 2024
- Selected for 2023 JuMP (Jacobs Mentoring Program)
- Extra Miler Award, April 2023

ASRC Federal • Huntsville, AL

2018 – 2019

Senior Database Engineer

Backend Database Engineer for two Missile Defense Data Center (MDDC) software applications, DMS and T-PATS / TREAD. Managed databases and developed SQL queries to create data reports. Enabled users to create, track, and manage data artifacts and to author Integrated Data Management Plan (IDMP) documentation.

Wavelink, Inc. • Huntsville, AL

2015 – 2018

Senior Software Engineer

Led effort to develop a Post-Flight Analysis Application (PFA), which used Java and NASA WorldWind to receive, process, and display flight analysis data messages in real time from a dynamic client. Defined a JSON-based data message format that maps to and expands the existing MIL-STD-1553 avionics data protocol messages, allowing human-readable client-server communication between the data source and the visualization application.

Supported front-end development for a CMMI-Level 4 global missile defense simulation software suite (Objective Simulation Framework), written in Java and C++ and used by the US and its allies worldwide for real-time HWIL and digital defense simulation.

Northrop Grumman Information Systems • Huntsville, AL

2011 – 2015

Software Engineer

Designed and built user interface prototypes for developmental IBCS (IAMD Battlespace Command System) and Ground Combat Vehicle (GCV) projects, using Adobe Creative Suite, jQuery, JavaScript, PHP, HTML5, and XML. Gathered feedback and conducted user interface prototype design tests to improve usability in Huntsville, AL; Minneapolis, MN; Fort Sill, OK; and Fort Benning, GA.

Worked large-scale IRAD for new Battlespace Command System capability marketed to foreign allies. Implemented Battlespace Definition algorithm for analysis using MATLAB, including sensor access, weapon access, and threat evaluation simulation modules in C++.

Won *Northrop Grumman Timely Award* (December 2013) for completing a future-concept GCV Armored Personnel Carrier simulation and user interface design prototype via the Unity3D game engine. The prototype tested soldiers' interaction with GUIs at Fort Benning, GA, for a conceptual touch-screen-driven GCV within scenario and environment. Designed and maintained team business development database with a web interface that prioritized and tracked stakeholder contacts from 65+ captures — internal, external, and potential projects worth \$4.8B+.

Teledyne Brown Engineering • Huntsville, AL

2006 – 2011

Software Developer

Designed and developed Adobe Flex Web 2.0 user interface prototypes for the IAMD CWMI (Common Warfighter Machine Interface) team. Integrated software simulation components in the PCIL (Prime Consolidated Integration Lab) real-time simulation laboratory at Boeing. Worked with logistics group to develop outlines and content for technical manuals. Trained users on new features.

Won 2 *TBE Presidential Awards for Outstanding Performance*: January 2010 and September 2010.

Helped plan and lead engineering and software development effort for a C# and WPF-based software prototype for a ruggedized, hand-held PC enabling mobile auditing for OSHA and environmental requirements.

Led 3-person team to develop, support, and maintain a modular database web portal that allowed users to track system anomalies encountered in testing. Web application written in Adobe ColdFusion with SQL, JavaScript, and HTML — used throughout the Boeing GMD Program.

EDUCATION

University of Alabama in Huntsville (UAH)

Huntsville, AL

Master of Science — Computer Science: Modeling & Simulation (GPA 3.45)

One of the first 2 graduates in a new course of study at UAH — one of only 3 graduate programs in Modeling and Simulation worldwide.

University of Alabama in Huntsville (UAH)

Huntsville, AL

Bachelor of Science — Computer Science (Minor: Mathematics)

Auburn University

Auburn, AL

Bachelor of Fine Arts — Visual Communication

PRESENTATIONS

- Segrest, F. M. Introduction to React Web Application Development. November 9, 2023. Marshall Space Flight Center: Huntsville, AL.
- Segrest, F. M. Streamlining DevOps using the Atlassian Jira Python API. November 2, 2023. Marshall Space Flight Center: Huntsville, AL.
- Segrest, F. M., Dodson, J. E., & Graham, M. Leveraging Game Development Engines to Simulate Conceptual Systems. March 29 – April 3, 2014. 26th Annual IEEE Software Technology Conference: Long Beach, CA.
- Segrest, F. M., Dodson, J. E., & Graham, M. Leveraging Game Development Engines to Simulate Conceptual Systems. April 7 – 10, 2014. Northrop Grumman Software Symposium: Baltimore, MD.

PUBLICATIONS

- Segrest, F. M. Streamlining DevOps with the Atlassian Jira Python API. Jacobs/NASA internal publication, 2024.
- Segrest, F. M. Coverage of Alabama Delegation at 2008 Republican National Convention. The Huntsville Times: September 2 – 5, 2008; al.com, September 1 – 5, 2008.
- Segrest, F. M. Open Source Telephony Answers the Call. Linux World Magazine: September 2005.
- Segrest, F. M. IAX Protocol Simplifies VoIP. Network World Magazine: April 11, 2005.
- Segrest, F. M., & Kervin, B. Say Hello to Asterisk. Linux Magazine: September 2004.